

Industrial Visit Report

Organization: PSG INDU TECH

Focus Area: Technical Textile Manufacturing & Testing

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1. Introduction

The industrial visit to PSG INDU TECH provided direct exposure to advanced technical textile manufacturing and testing practices. The facility operates as a research-driven centre integrating product development, testing, and innovation under standardized frameworks. The visit helped bridge theoretical knowledge with industrial-scale applications, particularly in functional textiles such as filtration, flame-retardant fabrics, and thermal protective materials.

2. Overview of PSG INDU TECH

PSG INDU TECH functions as a specialized centre focusing on:

- Development of technical textiles
- Advanced material testing
- Industry-oriented research projects
- Standard-based product validation

The facility is equipped with high-end instruments, some of which are unique and not commonly available in standard textile labs.

3. Exposure to Manufacturing Concepts

3.1 Technical Textile Structures Observed

- Nonwoven fabrics (Needle punching, thermal bonding)
- Coated textiles (single layer, multilayer, composite coatings)
- UV protective fabrics
- Composite materials (glass, acrylic-based)

3.2 Key Learning

Structure directly influences performance (filtration, insulation, protection).

Coating type determines durability and application life.

Composite materials enhance multifunctionality.

4. Testing and Evaluation Methods

4.1 Importance of Testing

- Ensures product reliability and safety
- Validates performance under real-life conditions
- Reduces failure in critical applications (automotive, protective wear)

4.2 Types of Testing Observed

a) Fire Testing

- Vertical flammability testing
- Horizontal flammability testing
- 45° inclined testing

Parameters measured:

1. After-flame time
2. After-glow time
3. Char length
4. Dripping behaviour

b) Thermal Conductivity Testing

- Measurement of heat transfer through materials
- Important for insulation and protective textiles

c) Filtration Testing

- Evaluation of air/liquid filtration efficiency
- Used in industrial and environmental applications

d) Wet and Analytical Testing

- Water resistance
- Surface characteristics
- Chemical behaviour

e) Advanced Instrumentation

- EDX (Elemental analysis)
- FE-SEM (Field Emission Scanning Electron Microscope)

Testing was conducted based on recognized standards, ensuring reliability and global acceptance.

Key Standards Observed:

- ASTM standards (e.g., flame resistance testing)
- ISO standards for environmental and conditioning requirements

Standardization Includes:

- Sample preparation
- Conditioning (temperature & humidity)
- Testing procedure
- Result interpretation

One of the most significant observations was the availability of specialized equipment that is rare and globally advanced.

These include:

- High-precision thermal testing instruments
- Advanced microscopy systems
- Large-capacity testing setups for industrial-scale fabrics

These facilities enable:

- Accurate research outcomes
- Industry-level product validation
- Innovation in technical textile solutions

7. Applications Observed

The tested and developed materials are used in:

- Filtration systems (industrial air/liquid filtration)
- Flame-retardant clothing (fire safety)
- Automotive textiles
- Protective textiles
- Thermal insulation materials

- Wipes and Masks
- Printing – Materials used Acrylic, wood etc...

8. Research and Problem-Solving Approach

A key takeaway from the visit was the structured approach to solving industrial problems:

- Identify real-world problem
- Select appropriate material and structure
- Apply standard testing methods
- Analyse results
- Optimize product performance

This approach emphasizes practical innovation backed by standards, not just theoretical development.

9. Personal Learning and Experience

- Gained clarity on how testing validates textile performance
- Understood the importance of standards in industrial applications
- Observed real-time research projects solving industrial challenges
- Learned how material structure, coating, and finishing affect properties
- Experienced exposure to advanced global-level equipment

This visit improved my understanding of:

- Technical textile applications
- Testing methodologies
- Industrial research workflow

10. Conclusion

The visit to PSG INDU TECH was highly informative and technically enriching. It provided practical insights into manufacturing, testing, and research in technical textiles. The exposure to advanced equipment, standardized testing, and real-world problem-solving approaches has strengthened my understanding and interest in technical textile development and innovation.